

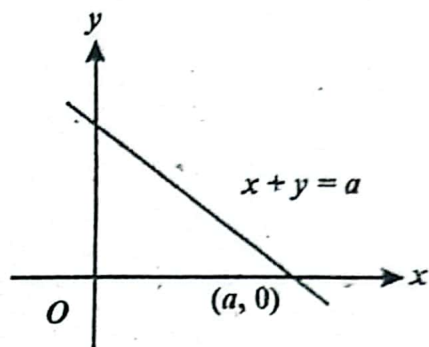
Exercise Set 10.2 (Page 468)

1. By means of double integration, find the area of the region bounded by

(a). the coordinate axes and the straight line $x + y = a$

Sol.(a). The required area

$$\begin{aligned}
 A &= \int_0^a \int_{a-x}^0 dy \, dx = \int_0^a [y]_{a-x}^0 dx \\
 &= \int_0^a (x-a) \, dx = \left[\frac{x^2}{2} \right]_0^a = \frac{a^2}{2}.
 \end{aligned}$$

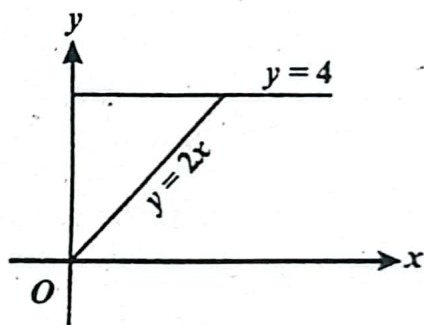


- 1(b). the y -axis, the straight line $y = 2x$, and the straight line $y = 4$

Sol.

(b). Required area

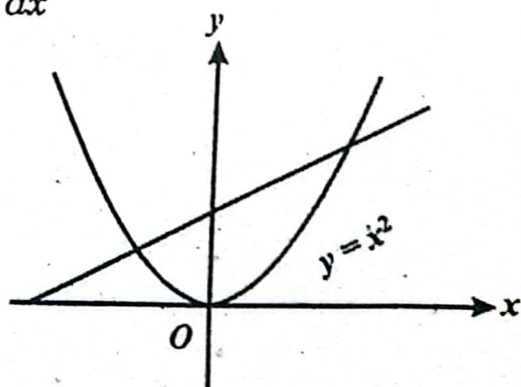
$$\begin{aligned}
 A &= \int_0^2 \int_{2x}^4 dy \, dx = \int_0^2 [y]_{2x}^4 dx \\
 &= \int_0^2 (4 - 2x) \, dx \\
 &= [4x - x^2]_0^2 = 8 - 4 = 4.
 \end{aligned}$$



2. Find the area bounded by the parabola $y = x^2$ and the straight line $y = 2x + 3$

Sol. Solving $y = x^2$ and $y = 2x + 3$ simultaneously, we get the limits of integration for x as $-1, 3$. The required area is

$$\begin{aligned}
 &= \int_{-1}^3 \int_{x^2}^{2x+3} dy \, dx = \int_{-1}^3 [y]_{x^2}^{2x+3} dx \\
 &= \int_{-1}^3 [2x + 3 - x^2] \, dx \quad \checkmark \\
 &= \left[x^2 + 3x - \frac{x^3}{3} \right]_{-1}^3 \\
 &= 9 + 9 - 9 - \left(1 - 3 + \frac{1}{3} \right) = \frac{32}{3}.
 \end{aligned}$$



3. Find the area bounded by $x^2 = 4y$ and $8y = x^2 + 16$

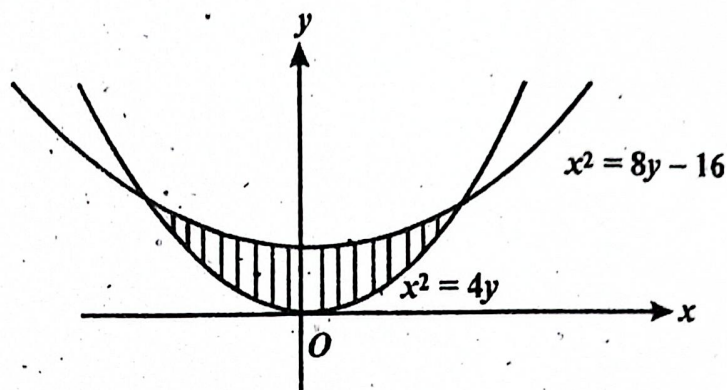
Sol. Solving the equation $x^2 = 4y$ and $8y = x^2 + 16$, we get $x = \pm 4$. The required area is

$$= \int_{-4}^4 \int_{\frac{x^2}{4}}^{\frac{x^2}{8} + 2} dy dx$$

$$= \int_{-4}^4 \left[y \right]_{\frac{x^2}{4}}^{\frac{x^2}{8} + 2} dx$$

$$= \int_{-4}^4 \left[\frac{x^2}{8} + 2 - \frac{x^2}{4} \right] dx = \int_{-4}^4 \left[2 - \frac{x^2}{8} \right] dx$$

$$= \left[2x - \frac{1}{8} \frac{x^3}{3} \right]_{-4}^4 = 8 - \frac{8}{3} - \left(-8 + \frac{8}{3} \right) = 16 - \frac{16}{3} = \frac{32}{3}.$$



4. Find the area outside the circle $r = 3$ and inside the cardioid $r = 2(1 + \cos \theta)$.

Sol. The curves intersect at $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$

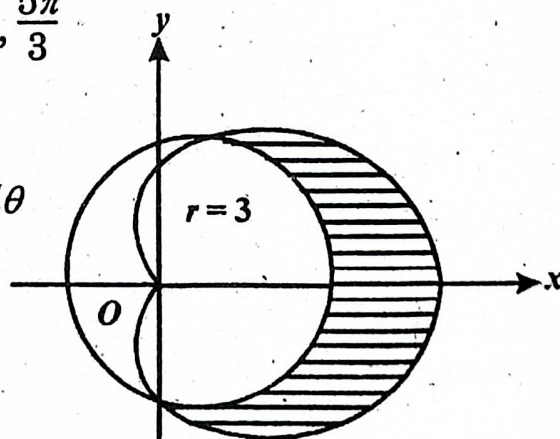
Required area

$$= 2 \int_0^{\pi/3} \int_3^{2(1+\cos\theta)} r dr d\theta$$

$$= 2 \int_0^{\pi/3} \left[\frac{r^2}{2} \right]_3^{2(1+\cos\theta)} d\theta$$

$$= 2 \int_0^{\pi/3} \left[\frac{4(1 + 2\cos\theta + \cos^2\theta)}{2} - \frac{9}{2} \right] d\theta$$

$$= \int_0^{\pi/3} (-5 + 8\cos\theta + 4\cos^2\theta) d\theta$$



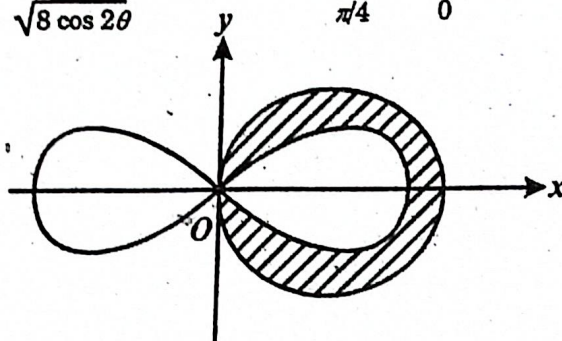
$$\begin{aligned}
 &= \int_0^{\pi/3} (-5 + 8 \cos \theta + 2(1 + \cos 2\theta)) d\theta \\
 &= [-3\theta + 8 \sin \theta + \sin 2\theta]_0^{\pi/3} \\
 &= -3 \cdot \frac{\pi}{3} + 8 \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \frac{9\sqrt{3}}{2} - \pi.
 \end{aligned}$$

5. Find the area inside the circle $r = 4 \sin \theta$ and outside the lemniscate $r^2 = 8 \cos 2\theta$.

Sol. Since the two curves intersect at $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$ and $r^2 = 8 \cos 2\theta = 0$ at

$\theta = \frac{\pi}{4}$, the desired area

$$= 2 \int_{\pi/6}^{\pi/4} \int_{\sqrt{8 \cos 2\theta}}^{4 \sin \theta} r dr d\theta + 2 \int_{\pi/4}^{\pi/2} \int_0^{4 \sin \theta} r dr d\theta$$



$$\begin{aligned}
 &= 2 \int_{\pi/6}^{\pi/4} \left[\frac{r^2}{2} \right]_{\sqrt{8 \cos 2\theta}}^{4 \sin \theta} d\theta + 2 \int_{\pi/4}^{\pi/2} \left[\frac{r^2}{2} \right]_0^{4 \sin \theta} d\theta \\
 &= 2 \int_{\pi/6}^{\pi/4} (8 \sin^2 \theta - 4 \cos 2\theta) d\theta + 2 \int_{\pi/4}^{\pi/2} 8 \sin^2 \theta d\theta \\
 &= 8 \int_{\pi/6}^{\pi/4} (1 - 2 \cos 2\theta) d\theta + 8 \int_{\pi/4}^{\pi/2} (1 - \cos 2\theta) d\theta \\
 &= 8 \left[\theta - \sin 2\theta \right]_{\pi/6}^{\pi/4} + 8 \left[\theta - \frac{\sin 2\theta}{2} \right]_{\pi/4}^{\pi/2} \\
 &= 8 \left[\frac{\pi}{4} - 1 - \frac{\pi}{6} + \frac{\sqrt{3}}{2} \right] + 8 \left[\frac{\pi}{2} - \frac{\pi}{4} + \frac{1}{2} \right] \\
 &= 8 \left[\frac{3\pi - 12 - 2\pi + 6\sqrt{3}}{12} \right] + 8 \left[\frac{\pi}{4} + \frac{1}{2} \right]
 \end{aligned}$$

$$= \frac{2\pi}{3} + 4\sqrt{3} - 8 + 2\pi + 4 = \frac{8\pi}{3} + 4\sqrt{3} - 4.$$

6. Find the volume in the first octant between the planes $z = 0$, $z = x + y + 2$ and inside the cylinder $x^2 + y^2 = 16$

$$\begin{aligned} \text{Sol. } V &= \int_0^4 \int_0^{\sqrt{16-x^2}} (x+y+2) dy dx \\ &= \int_0^4 \left[xy + \frac{y^2}{2} + 2y \right]_0^{\sqrt{16-x^2}} dx \\ &= \int_0^4 \left[x\sqrt{16-x^2} + \frac{16-x^2}{2} + 2\sqrt{16-x^2} \right] dx \\ &= -\frac{1}{3} [(16-x^2)^{3/2}]_0^4 + \left[8x - \frac{x^3}{6} \right]_0^4 + 2 \left[\frac{x\sqrt{16-x^2}}{2} + 8 \arcsin \frac{x}{4} \right]_0^4 \\ &= \frac{64}{3} + \left(32 - \frac{64}{6} \right) + 2 \left[8 \left(\frac{\pi}{2} \right) \right] = \frac{64}{3} + \frac{64}{3} + 8\pi = \frac{128}{3} + 8\pi. \end{aligned}$$

7. Find the volume bounded by the cylinder $x^2 + y^2 = 4$ and the planes $y + z = 4$ and $z = 0$

$$\begin{aligned} \text{Sol. } V &= \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (4-y) dy dx = 2 \int_{-2}^2 \int_0^{\sqrt{4-x^2}} 4 dy dx \\ &= 8 \int_0^2 \sqrt{4-x^2} dx = 16 \int_0^2 \sqrt{4-x^2} dx \\ &= 16 \left[x \frac{\sqrt{4-x^2}}{2} + 2 \arcsin \frac{x}{2} \right]_0^2 = 16 \left(2 \cdot \frac{\pi}{2} \right) = 16\pi. \end{aligned}$$

8. Find the volume of the solid in the first octant bounded by the coordinate planes and the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$, a , b and c being positive.

$$\text{Sol. From } \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1, \text{ we have } z = c \left(1 - \frac{x}{a} - \frac{y}{b} \right).$$

Volume of the solid

$$\begin{aligned}
&= \int_0^a \int_0^{b(1-\frac{x}{a})} c \left(1 - \frac{x}{a} - \frac{y}{b} \right) dy dx \\
&= c \int_0^a \left[y - \frac{xy}{a} - \frac{y^2}{2b} \right]_0^{b(1-\frac{x}{a})} dx \\
&= c \int_0^a \left[b \left(1 - \frac{x}{a} \right) - \frac{x}{a} b \left(1 - \frac{x}{a} \right) - \frac{1}{2b} \cdot b^2 \left(1 - \frac{x}{a} \right)^2 \right] dx \\
&= c \int_0^a \left[b - \frac{bx}{a} - \frac{bx}{a} + b \frac{x^2}{a^2} - \frac{b}{2} \left(1 - \frac{2x}{a} + \frac{x^2}{a^2} \right) \right] dx \\
&= c \int_0^a \left(\frac{b}{2} - \frac{bx}{a} + \frac{bx^2}{2a^2} \right) dx = c \left[\frac{bx}{2} - \frac{b}{a} \frac{x^2}{2} + \frac{b}{2} \frac{x^3}{3a^2} \right]_0^a \\
&= c \left[\frac{ba}{2} - \frac{ba}{2} + \frac{ba}{6} \right] = \frac{abc}{6}
\end{aligned}$$

9. Find the volume of the solid bounded by the paraboloid $z = 4 - x^2 - y^2$ and the xy -plane.

sol. The region D in the xy -plane is bounded by $x^2 + y^2 = 4$.

$$\begin{aligned}
\text{Volume} &= \iint_D (4 - x^2 - y^2) dx dy = \int_0^{2\pi} \int_0^2 (4 - r^2) r dr d\theta \\
&= \int_0^{2\pi} \left[2r^2 - \frac{r^4}{4} \right]_0^2 d\theta = 4 \int_0^{2\pi} d\theta = 8\pi.
\end{aligned}$$

10. Find the volume of the solid bounded by the graphs of $x^2 + y^2 = 4$, $z = \sqrt{16 - x^2 - y^2}$, $z = 0$.

sol. Required volume

$$V = \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \sqrt{16 - x^2 - y^2} dy dx$$

We change the integral into polar coordinates. From $x = r \cos \theta$, $y = r \sin \theta$, we have $x^2 + y^2 = r^2$, so that

$z = \sqrt{16 - r^2}$. The bounds become $0 \leq r \leq 2$ and $0 \leq \theta \leq 2\pi$

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^2 \sqrt{16 - r^2} r dr d\theta = \frac{1}{3} \int_0^{2\pi} [(16 - r^2)^{3/2}]_0^2 d\theta \\ &= -\frac{1}{3} \int_0^{2\pi} [(12)^{3/2} - (16)^{3/2}] d\theta = -\frac{1}{3} \int_0^{2\pi} (24\sqrt{3} - 64) d\theta \\ &= -\frac{1}{3} (24\sqrt{3} - 64) \theta \Big|_0^{2\pi} = -\frac{1}{3} (24\sqrt{3} - 64) 2\pi \\ &= \frac{-16\pi}{3} (3\sqrt{3} - 8) = \frac{16\pi}{3} (8 - 3\sqrt{3}). \end{aligned}$$

11. Find the centre of gravity of the plane area of uniform density bounded by

$$x^2 = 4y \text{ and } 8y = x^2 + 16 \text{ in the first quadrant.}$$

Sol. The parabolas intersect at the points where

$$\frac{x^2}{4} = \frac{x^2 + 16}{8}, \quad \text{or} \quad x = \pm 4$$

Points of intersection are $(4, 4)$, $(-4, 4)$. Let $(\bar{x}, \bar{y})$ be coordinates of the centre of gravity of the area lying in the first quadrant. Then

$$\bar{x} = \frac{\int \int x dA}{\int \int dA}; \quad \bar{y} = \frac{\int \int y dA}{\int \int dA}$$

$$\begin{aligned} \text{Now, } \int \int dA &= \int_0^4 \int_{x^2/4}^{(x^2+16)/8} dy dx = \int_0^4 \left(\frac{x^2 + 16}{8} - \frac{x^2}{4} \right) dx \\ &= \frac{1}{8} \int_0^4 (16 - x^2) dx = \frac{1}{8} \left[16x - \frac{x^3}{3} \right]_0^4 = \frac{16}{3}. \end{aligned}$$

$$\begin{aligned} \int \int x dA &= \int_0^4 \int_{x^2/4}^{(x^2+16)/8} x dy dx = \frac{1}{8} \int_0^4 (16x - x^3) dx \\ &= \frac{1}{8} \left[8x^2 - \frac{x^4}{4} \right]_0^4 = 8. \end{aligned}$$

$$\text{Hence } \bar{x} = \frac{\frac{8}{15}}{\frac{16}{3}} = \frac{3}{2}$$

$$\iint y \, dA = \int_0^4 \int_{x^2/4}^{(x^2+16)/8} y \, dy \, dx$$

$$= \frac{1}{2} \int_0^4 \left[\left(\frac{x^2+16}{8} \right)^2 - \left(\frac{x^2}{4} \right)^2 \right] dx$$

$$= \frac{1}{2} \cdot \frac{1}{64} \int_0^4 (256 + 32x^2 - 3x^4) \, dx$$

$$= \frac{1}{128} \left[256x + \frac{32x^3}{3} - \frac{3x^5}{5} \right]_0^4$$

$$= \frac{1}{128} \cdot 256 \times 4 \left[1 + \frac{2}{3} - \frac{3}{5} \right] = \frac{8 \times 16}{15}$$

$$\bar{y} = \frac{128/15}{16/3} = \frac{8}{5}$$

$$\text{Centre of gravity } (\bar{x}, \bar{y}) = \left(\frac{3}{2}, \frac{8}{5} \right).$$

